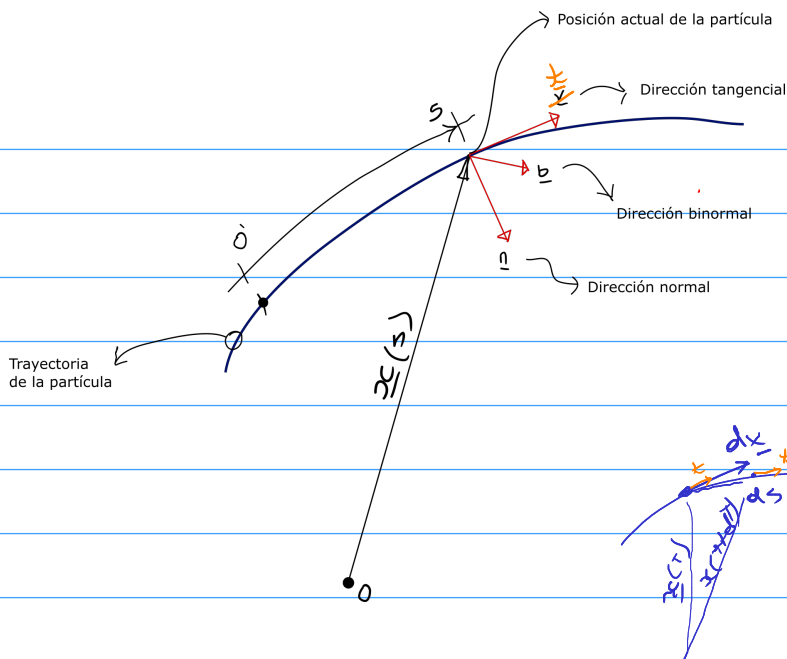


$$\underline{x} = \begin{bmatrix} x \\ y \end{bmatrix}$$

$$\begin{matrix} x(t) \\ y(t) \end{matrix}$$

$$\underline{t} = \frac{d\underline{x}}{ds} = \frac{d\underline{x}/dt}{ds/dt} \Rightarrow \underline{t} = \frac{\underline{v}}{\|\underline{v}\|}$$

$$\underline{x}(T) = \underline{x}^*(s(T))$$

$$\underline{v} = \frac{d\underline{x}(T)}{dT} = \frac{d\underline{x}^*(s)}{ds} \cdot \frac{ds(T)}{dT} = \underline{v}(T) \cdot \underline{t}(s)$$

$$\underline{a} = \frac{d\underline{v}(T)}{dT} = \underbrace{\frac{dv(T)}{dT}}_{a_T} \cdot \underline{t}(s) + v(T) \cdot \frac{d\underline{t}(s)}{ds} \cdot \frac{ds(T)}{dT}$$

$$\underline{a} = a_T \underline{t}(s) + v(T)^2 \cdot k \underline{n}(s)$$

$$a_T = 0$$

$$\underline{a} = v(T)^2 \cdot \frac{r}{r^2 + \left(\frac{h}{2\pi}\right)^2} \cdot \underline{n}(s)$$

• $s(T) \rightarrow$ longitud de recorrido de la partícula

• $\frac{ds(T)}{dT} = v > 0 \quad v = \|\underline{v}\|$

• $\underline{t} = \frac{d\underline{x}(s)}{ds} \quad \|\underline{t}\| = 1$

• $\underline{n} = \frac{\frac{d\underline{t}}{ds}}{\left\|\frac{d\underline{t}}{ds}\right\|} = \frac{1}{k} \frac{d\underline{t}}{ds}$
curvatura

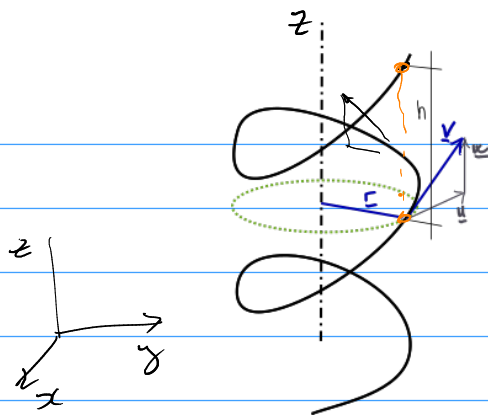
$$\|\underline{t}\|^2 = \underline{t} \cdot \underline{t} = 1$$

$$\frac{\partial}{\partial s}(\underline{t} \cdot \underline{t}) = \frac{\partial}{\partial s}(1)$$

$$2 \frac{\partial \underline{t}}{\partial s} \cdot \underline{t} = 0 \Rightarrow \frac{\partial \underline{t}}{\partial s} \perp \underline{t}$$

• $k = \frac{r}{r^2 + \left(\frac{h}{2\pi}\right)^2}$

• $\underline{b} = \underline{t} \times \underline{n}$



$$|\underline{v}| = v$$

$$|\underline{\omega}| = \omega$$

$$|\underline{u}| = u$$

$$\left. \begin{aligned} T_v &= \frac{2\pi r}{u} \\ h &= \omega \cdot T_v \end{aligned} \right\} \frac{h}{\omega} = \frac{2\pi r}{u} \Rightarrow \omega = \frac{h}{2\pi r} u$$

$$v^2 = \omega^2 + u^2 = \left[\left(\frac{h}{2\pi} \right)^2 + r^2 \right] \cdot \frac{u^2}{r^2}$$

$$\left[\frac{a}{r} = \left[\left(\frac{h}{2\pi} \right)^2 + r^2 \right] \frac{u^2}{r^2} \cdot \frac{r}{r^2 + \left(\frac{h}{2\pi} \right)^2} \cdot \frac{1}{r} = \frac{u^2}{r} \cdot \frac{1}{r} \right]$$

↓
Acceleration or centripetal